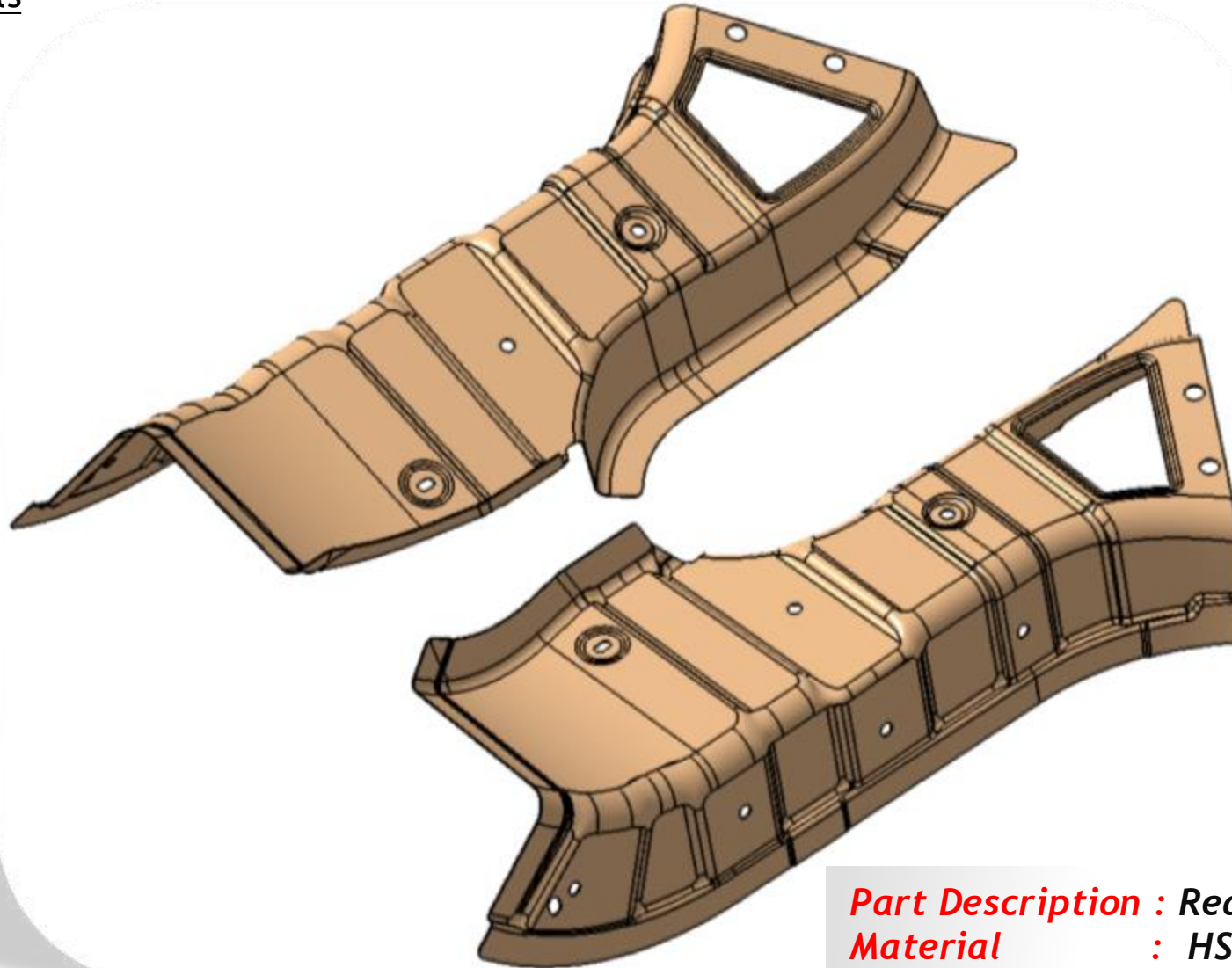


Part Image & Details



Part Description : Rear D Pillar Lower LH/RH
Material : HSLA 340
Thickness : 0.8mm

Part Geometry & Forming Challenges

✓ **620 × 220 × 130 mm → 68 mm Deep Flange → Tipping-Sensitive Geometry**

Part Geometry

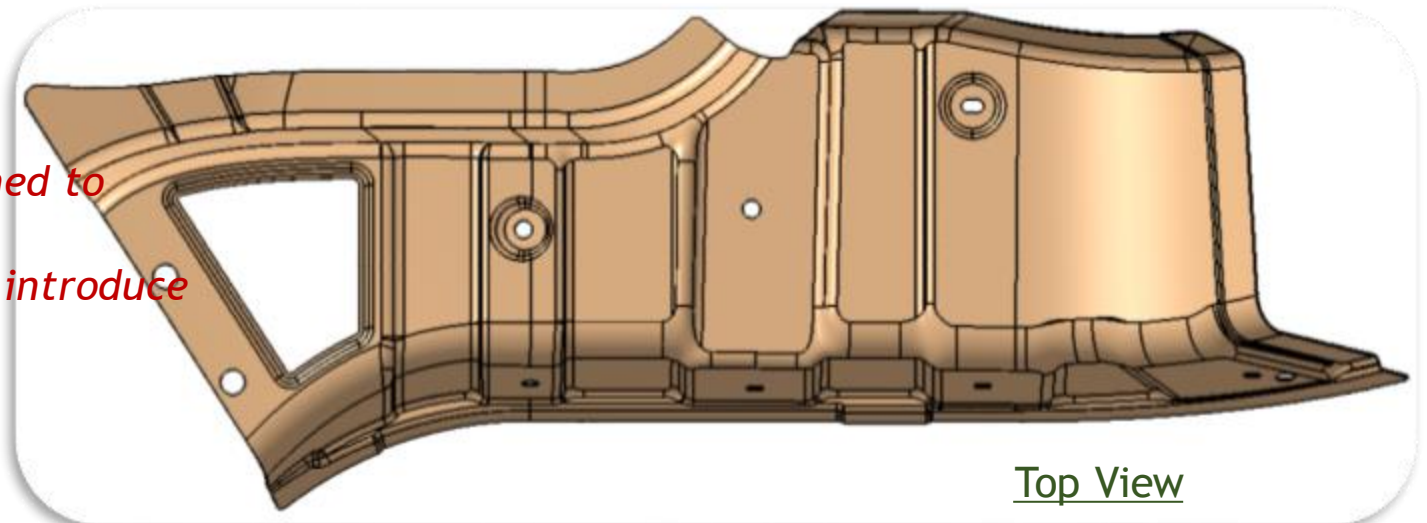
- Overall Size: **620 × 220 × 130 mm**
- Maximum Flange Depth: **~68 mm**
- Long, deep profile with varying flange and wall geometry.



Isometric View

Key Forming Challenge

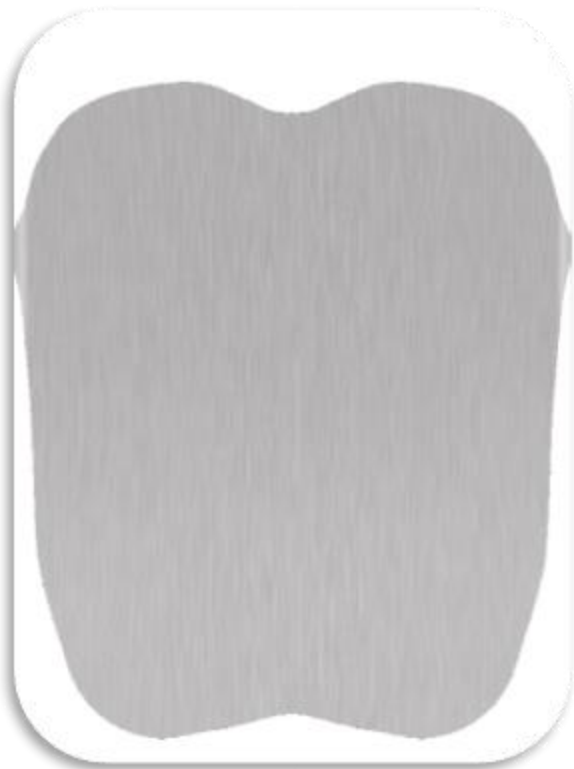
- Highly tipping-sensitive geometry.
- Tipping direction must be precisely defined to maintain a feasible forming direction.
- Even a small change in tipping angle can introduce local undercut conditions.



Top View

Operation sequences

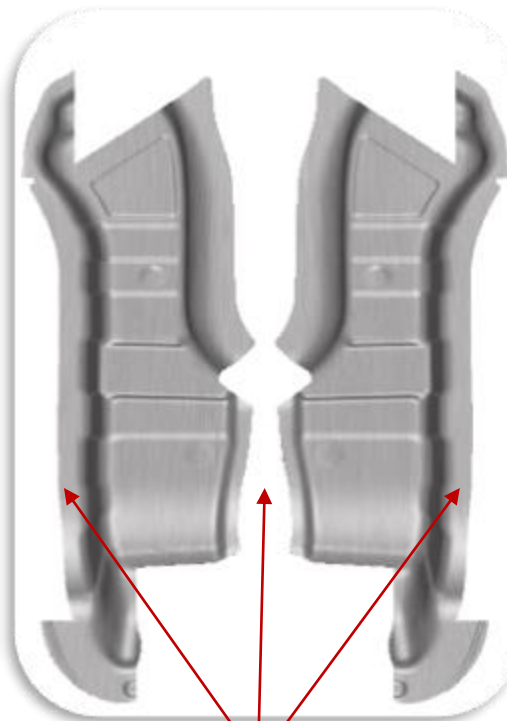
Process



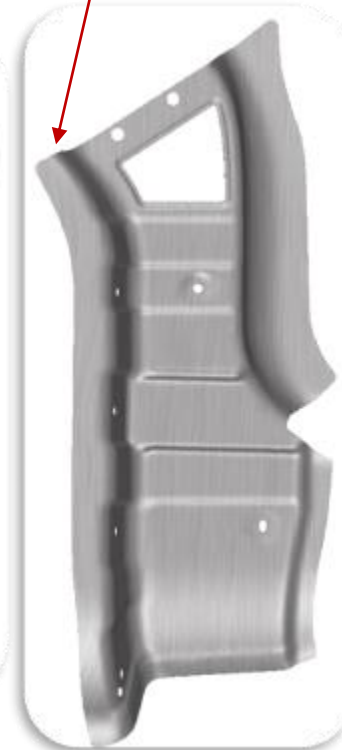
OP05-Profile Blank



OP10-Draw

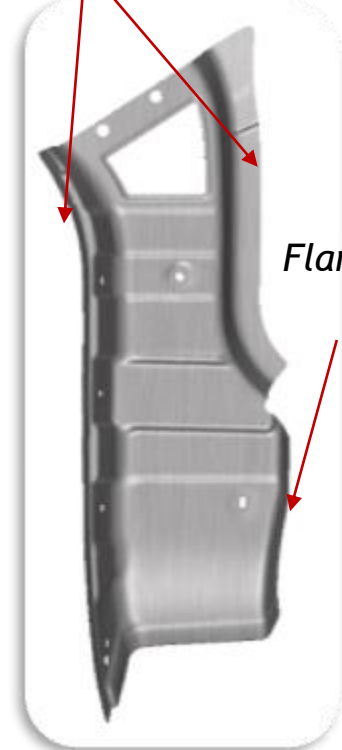


OP20-Part Off
&
Trim



Cam Trim

OP30-Pierce,
Cam Pierce &
Cam Trim



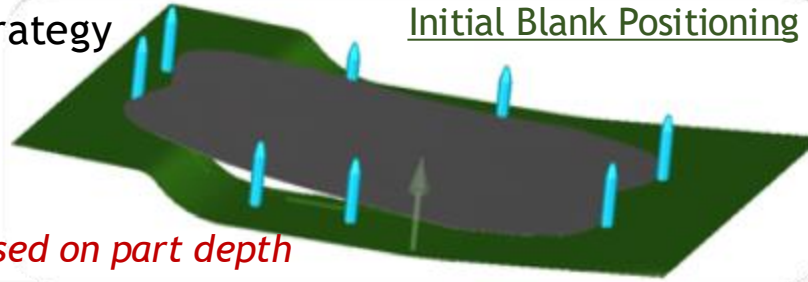
Flange Down

Flange Up

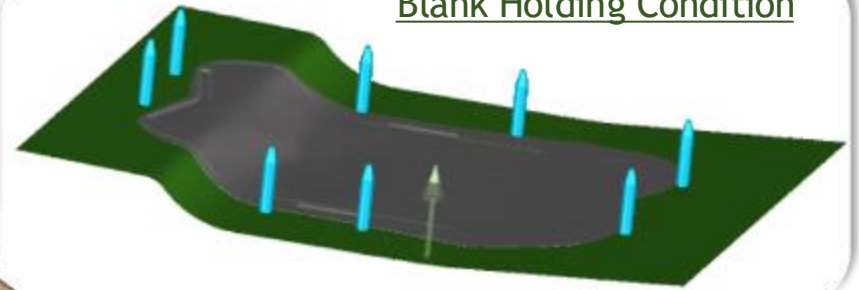
OP40-Flange
Down & up

Draw Process & Material Flow Strategy

Initial Blank Positioning



Blank Holding Condition



Binder Profile Strategy

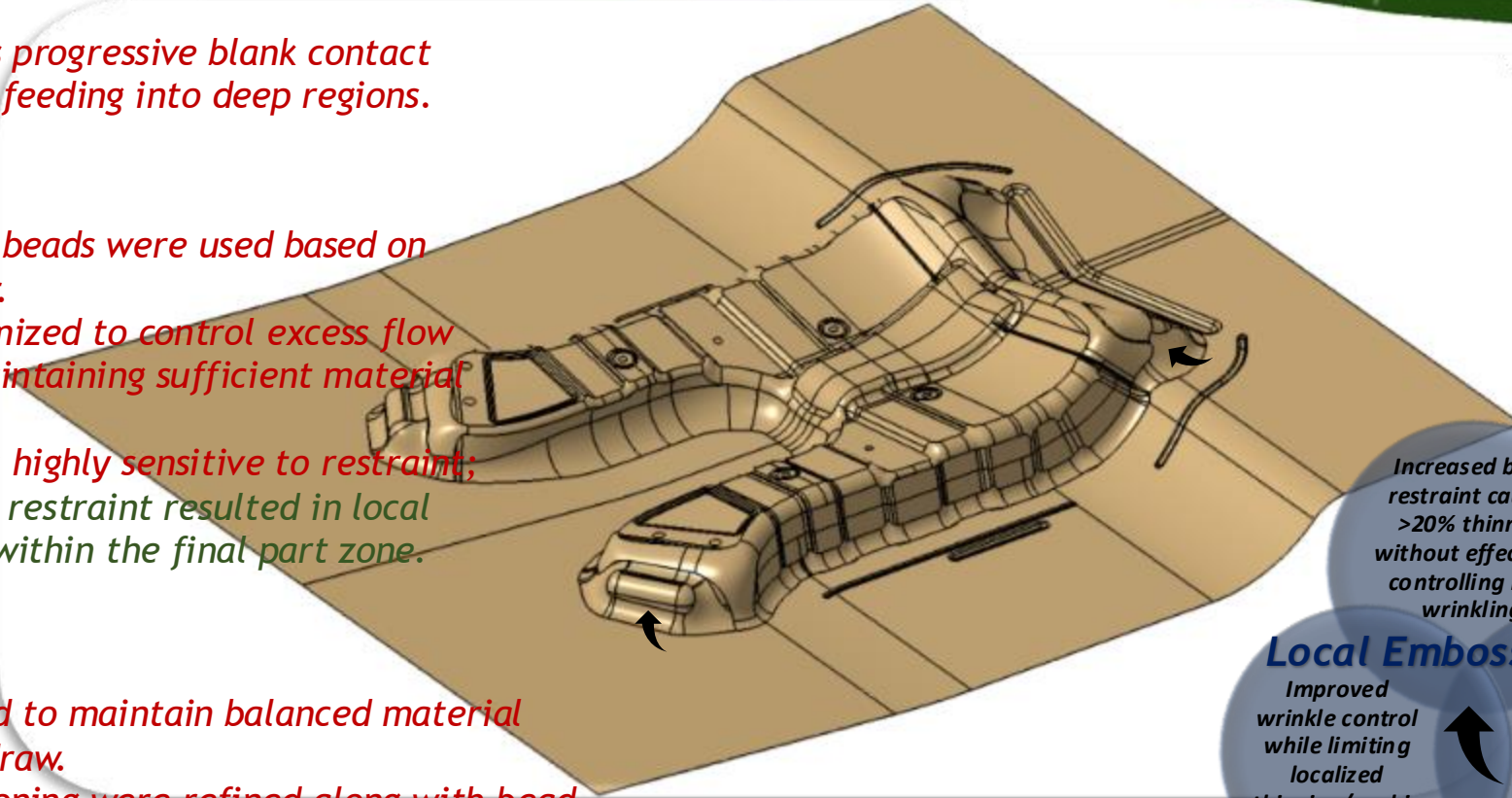
- Binder height/profile defined based on part depth and geometry variation.
- Height variation enables progressive blank contact and controlled material feeding into deep regions.

Draw Bead Strategy

- Only required/localized beads were used based on material-flow behaviour.
- Bead restraint was optimized to control excess flow and thickening while maintaining sufficient material feed.
- The forming condition is highly sensitive to restraint; even a slight increase in restraint resulted in local thinning exceeding 20% within the final part zone.

Blank Positioning

- Blank position optimized to maintain balanced material availability across the draw.
- Blank profile and positioning were refined along with bead restraint to achieve a stable material-flow condition.



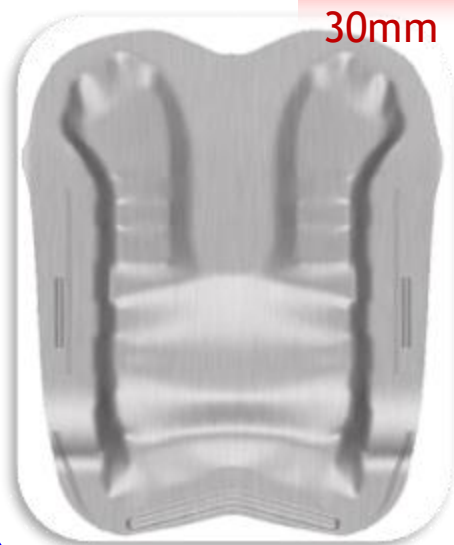
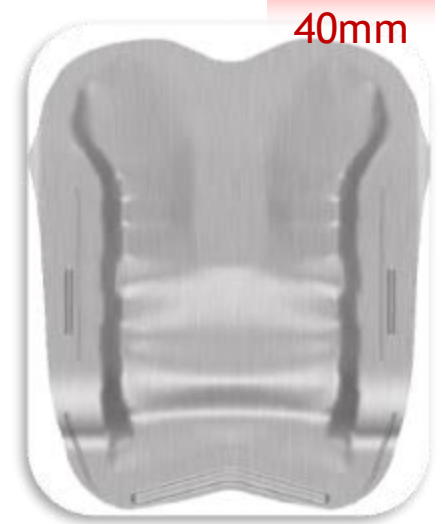
Increased bead restraint caused >20% thinning without effectively controlling local wrinkling.

Local Emboss Strategy

Improved wrinkle control while limiting localized thinning/necking.

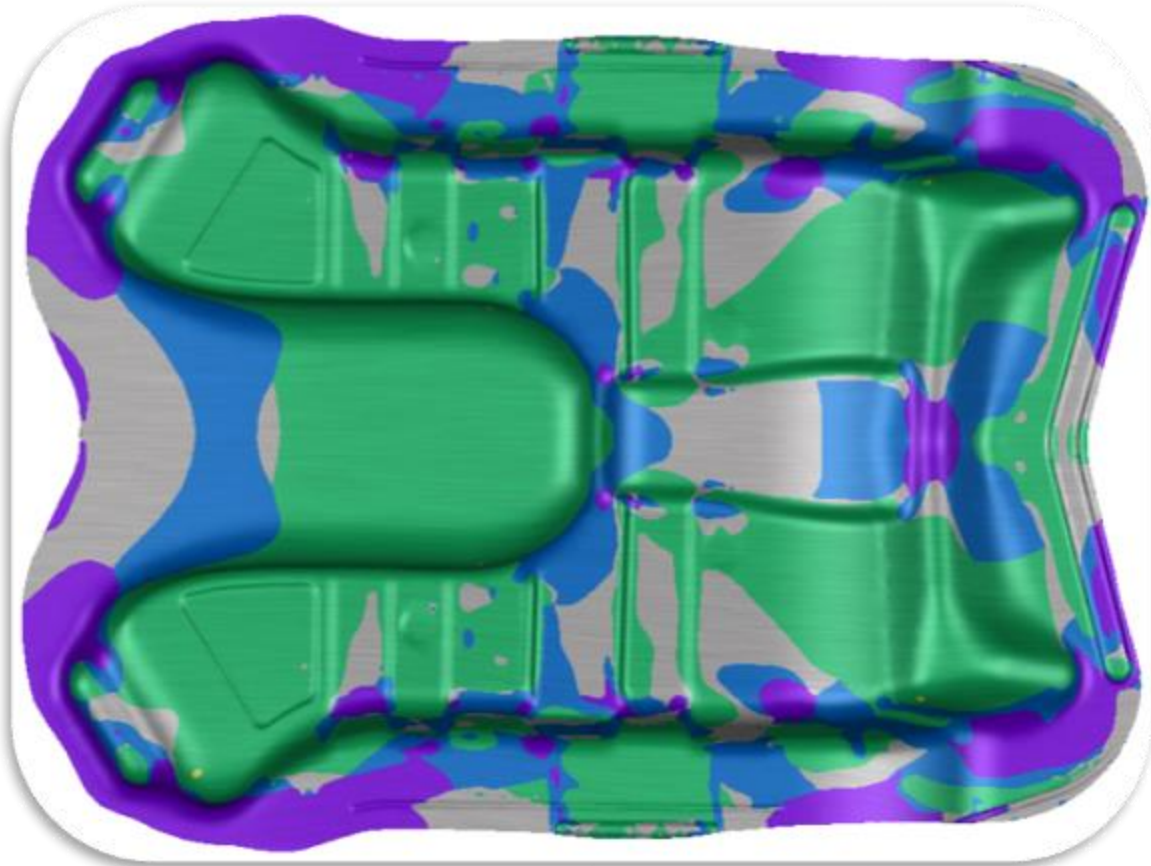
Local emboss introduced to manage excess material and redistribute strain.

Step-by-Step Draw Forming Sequence

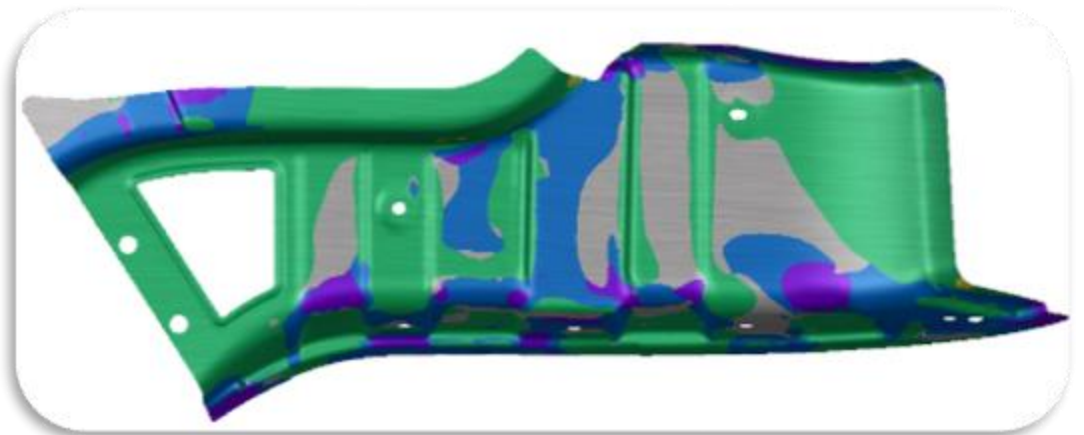
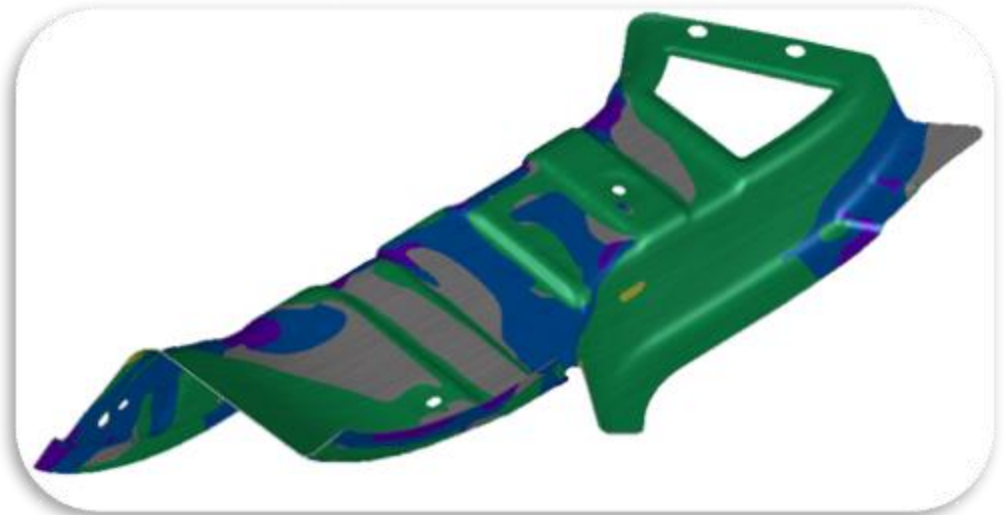


Final Part

FLD Analysis



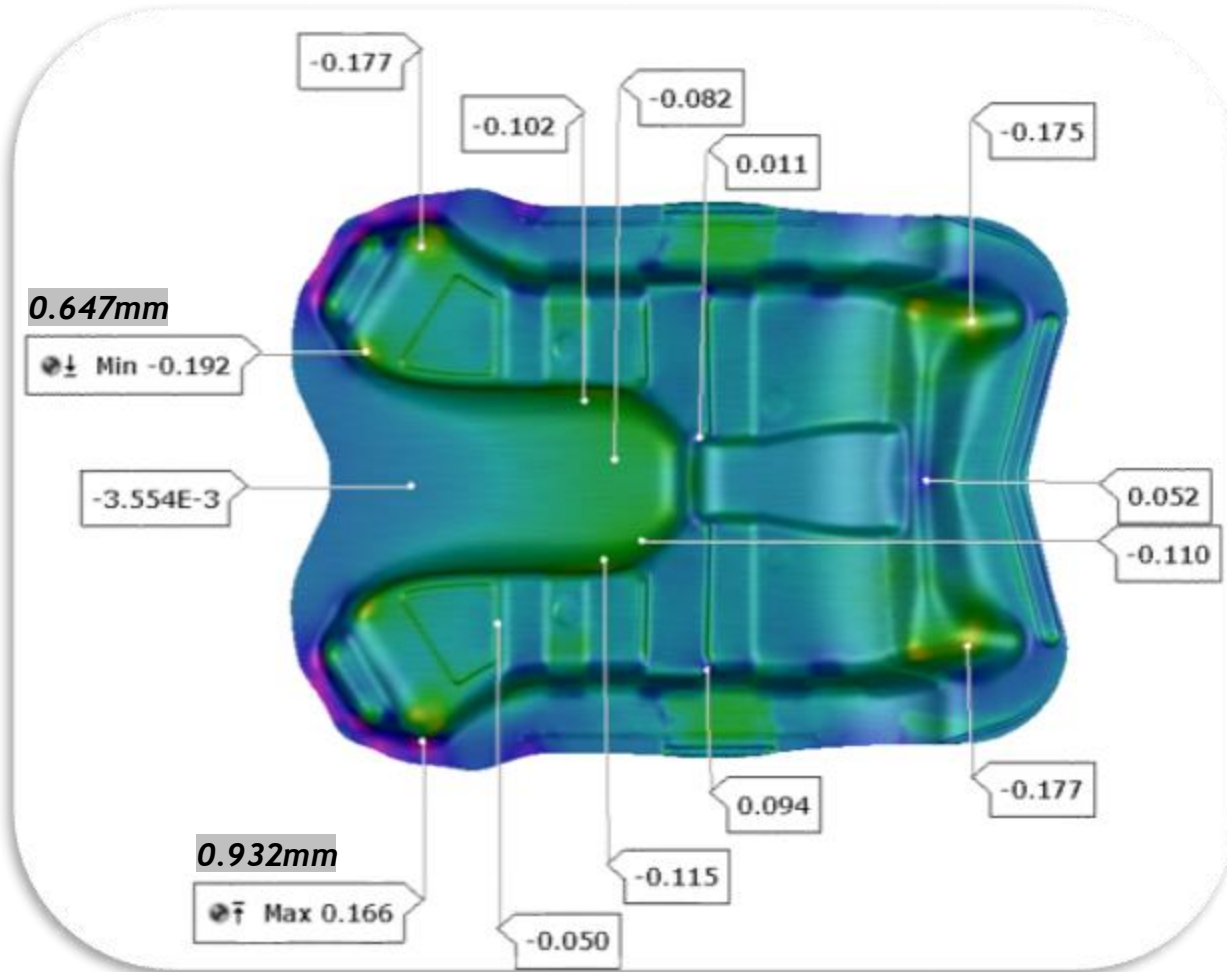
Draw



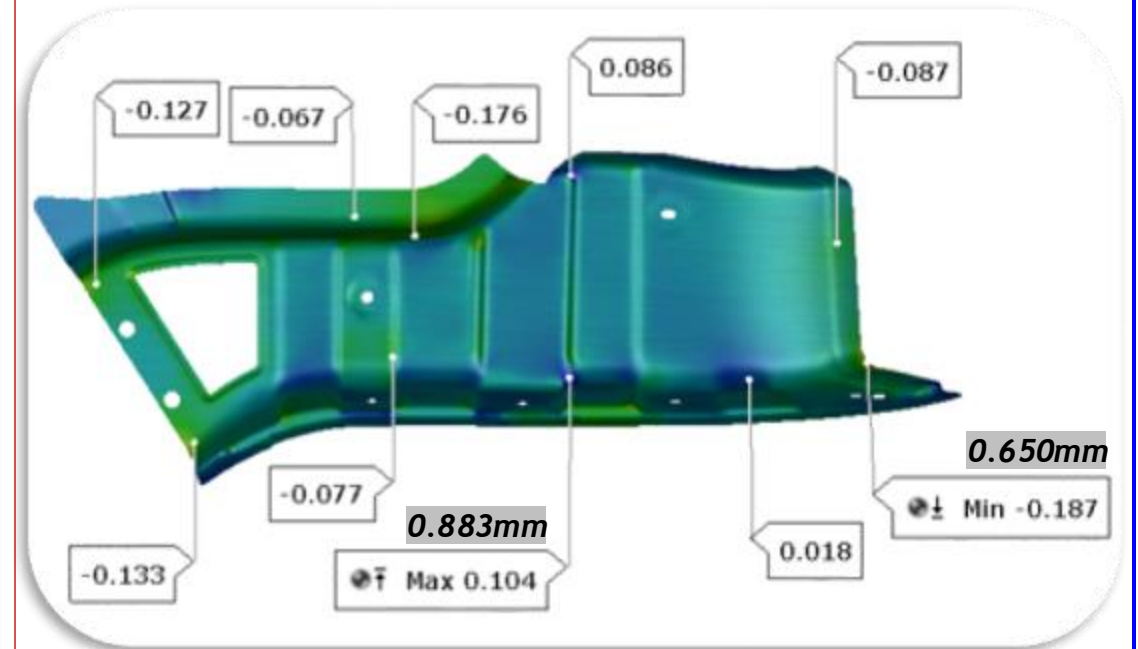
Final Part

Final Part

Thinning & Thickening



Draw



Final Part